The viscous strain bound: what a ν, t-dependent constant can and cannot give

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Lean record: `PeriodicNSContinuation/ViscousStrainBound.lean` (Apache-2.0) in `github.com/BenFrohman/periodic-ns-classical-continuation`.
This document does not claim a solution of the Navier–Stokes Millennium Problem.

Abstract

The uniform bound $\|S\|_\infty \le C\|\omega\|_\infty$ is false (Bahouri–Chemin family, Zenodo 10.5281/zenodo.23904623). We examine the surviving variant with a constant depending on ν and $t > 0$. We separate (i) a classical log-corrected bound, provable by standard Littlewood–Paley and parabolic-smoothing arguments but carrying a solution-dependent logarithm; (ii) the pure form $\|S(t)\|_\infty \le c(\nu, t)\|\omega(t)\|_\infty$, which is open; and (iii) the consequence of (ii), proved in Lean: it yields only a lifespan (small-data) bound, and a bounded constant does not exclude blow-up of the comparison ODE. So the variant, even if true, reduces back to the Beale–Kato–Majda criterion. Numerics for the evolved family (2D slice, full runs) show the ratio follows $a + b\log(1/(\nu t))$ with b close to the heat-flow value $1/\pi$.

1 Statements

Let u be a smooth solution on $\mathbb{T}^3 \times [0, T^*)$, $\omega = \nabla \times u$, $S = \frac{1}{2}(\nabla u + \nabla u^\top)$, $M(t) = \|\omega(t)\|_\infty$.

(V1) Log-corrected bound (classical). For $\alpha \in (0, 1)$ there is C_α with

$$\|S\|_\infty \le C_\alpha \|\omega\|_\infty \Big(1 + \log^+ \frac{\|\omega\|_{C^\alpha}}{\|\omega\|_\infty}\Big)$$

for mean-zero ω on $\mathbb{T}^3$ (dyadic splitting: low blocks $\lesssim \|\omega\|_\infty$ each, high blocks $\lesssim 2^{-j\alpha}\|\omega\|_{C^\alpha}$; cf. Beale–Kato–Majda, Kozono–Taniuchi, Bahouri–Chemin–Danchin). *Status: classical; not formalized here.*

(V2) Smoothing. Classical mild-solution theory gives, for $\tau \le \min(t/2, c_0/\bar{M})$ with $\bar{M} = \sup_{[t-\tau,t]} M$, $\|\omega(t)\|_{C^\alpha} \le B(\nu\tau)^{-\alpha/2}\bar{M}$. The nonlinear time scale $c_0/\bar{M}$ is forced: vorticity has units of inverse time. Combining with (V1) and the identity

$$\log\big(B(\nu\tau)^{-\alpha/2}R\big) = \log B - \tfrac{\alpha}{2}\log(\nu\tau) + \log R \qquad (\texttt{log_smoothing_split},\ \text{Lean})$$

gives

$$\|S(t)\|_\infty \le C\, M(t)\Big(1 + \log^+ \frac{\bar{M}}{M(t)} + \log^+ \frac{1}{\nu t} + \log^+ \frac{\bar{M}}{\nu}\Big).$$

Status: classical modulo constants (not tracked, not formalized). The constant depends on the solution through $\bar{M}/\nu$ and $\bar{M}/M(t)$; it is *not* a function of (ν, t) alone. Inserted in $D^+M \le \|S\|_\infty M$ it gives $M' \lesssim M^2 \log(e + M/\nu + 1/(\nu t))$, which does not close; this is exactly the BKM situation.

(V3) Pure viscous form (`ViscousStrainBound`). There is $c(\nu, t)$, finite for $t > 0$ and independent of the solution, with $\|S(t)\|_\infty \le c(\nu, t)M(t)$. *Status: OPEN.* Heat smoothing alone does not give it, because it controls $\|\omega(t)\|_{C^\alpha}$ by ω at earlier times, not by $M(t)$, and (V2) shows the natural constant depends on $\bar{M}/\nu$.

2 What (V3) would give (proved in Lean)

Definition 1. `VorticityMaxRiccati`: $M > 0$, $M' = m$, $m \le \|S\|_\infty M$ on $[t_0, T]$ (an idealization of the classical Dini-derivative inequality). `ViscousStrainBound`: $\|S(t)\|_\infty \le c(t)M(t)$ on $[t_0, T]$.

Theorem 1 (`riccati_comparison`, `bound_of_viscousStrainBound`)**.** *Under both hypotheses, with $C' = c$, the function $1/M + C$ is nondecreasing on $[t_0, T]$, and whenever $M(t_0) \int_{t_0}^t c < 1$,*

$$M(t) \le \frac{M(t_0)}{1 - M(t_0) \int_{t_0}^t c}.$$

Theorem 2 (`riccati_saturates`, `riccati_saturates_unbounded`)**.** *$M(t) = 1/(1-t)$ satisfies $M' = 1 \cdot M^2$ on $t < 1$ and $M \to \infty$ as $t \to 1^-$.*

So a constant $c(\nu, t)$, even bounded and integrable, gives a lifespan lower bound, not regularity. Regularity still requires $\int_0^{T^*} M < \infty$ (BKM), i.e. part (A) of the Frohmanian Tether Hypothesis or an equivalent input. **Verdict: (V1),(V2) classical; (V3) open; (V3)⇒ regularity is false at the ODE level, so the route reduces to BKM.**

3 Numerics (full runs only; 2D slice $u_3 = 0$, so no vortex stretching)

Data $\omega = A\,\omega_K$, pseudo-spectral, integrating-factor RK4, 2/3 dealiasing, $N = \max(256, 4K)$ (amplitude runs $N = 512$), 26 log-spaced times on $[0, T]$. We fit ratio $= a + b\log(1/(\nu t))$ over times with $\nu t K^2 > 1$; "heat" is the same ratio for $e^{\nu t\Delta}\omega_K$. The heat prediction is $\|S\|_\infty \approx \frac{2}{\pi}\log(1/\sqrt{\nu t})\,M$, i.e. $b = 1/\pi \approx 0.318$. Rows with too few fit points are marked –.

K	ν	n	a	b	b_{heat}
127	0.001	6	0.190	0.296	0.259
127	0.01	13	0.169	0.302	0.295
127	0.1	19	0.112	0.311	0.310
255	0.001	10	0.092	0.312	0.288
255	0.01	17	0.151	0.306	0.302
255	0.1	23	0.109	0.312	0.312
31	0.001	0	–	–	–
31	0.01	5	0.399	0.254	0.233
31	0.1	11	0.126	0.305	0.304
63	0.001	2	–	–	–
63	0.01	9	0.219	0.292	0.282
63	0.1	15	0.115	0.309	0.308

A	t	$M(t)$	ratio	heat ratio
1	0.0000	1.391	1.825	1.825
1	0.0022	1.380	1.833	1.833
1	0.0500	1.195	1.974	1.972
16	0.0000	22.252	1.825	1.825
16	0.0022	21.968	1.841	1.833
16	0.0500	18.395	1.770	1.972
32	0.0000	44.504	1.825	1.825
32	0.0022	44.140	1.829	1.833
32	0.0500	36.562	1.578	1.972
4	0.0000	5.563	1.825	1.825
4	0.0022	5.517	1.834	1.833
4	0.0500	4.759	1.953	1.972

Second table: $K = 31$, $\nu = 0.01$, $T = 0.05$, amplitudes $A = 1, 4, 16, 32$. Larger amplitude *lowers* the ratio in this window; no Reynolds-number growth of the ratio was seen. This is one 2D family over a short window and says nothing about near-singular 3D flows, which is exactly where (V3) is in question.

4 Caveats

2D slice only; one data family; grid sup of $\|S\|_{op}$; the fit is empirical. (V1),(V2) are cited, not formalized. The Lean statements are abstract scalar comparisons and are not assumed for Navier–Stokes anywhere.